微分几何习题参考解答*

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November 2023

1 第五章二部分习题

6.（注意本题描述有误，详见勘误）

$$\mathbf{e}_1 = \frac{\mathbf{r}_1}{\sqrt{E}}, \quad \mathbf{e}_2 = \frac{E\mathbf{r}_2 - F\mathbf{r}_1}{\sqrt{E(EG-F^2)}},$$

$$d\mathbf{r} = \mathbf{r}_\alpha du^\alpha = (\sqrt{E}du^1 + \frac{F}{\sqrt{E}}du^2)\mathbf{e}_1 + \sqrt{\frac{EG-F^2}{E}}du^2\mathbf{e}_2,$$

由 $d\mathbf{r} = \omega^\alpha e_\alpha$，得

$$\omega^1 = \sqrt{E}du^1 + \frac{F}{\sqrt{E}}du^2$$
$$\omega^2 = \sqrt{\frac{EG-F^2}{E}}du^2.$$

根据

$$\begin{cases} d\omega^1 = \omega^{12} \wedge \omega^2, \\ d\omega^2 = \omega^{21} \wedge \omega^1, \\ \omega^{12} = \omega^{21}, \end{cases}$$

*彭家贵、陈卿著《微分几何》，高等教育出版社 2021 年 11 月第 2 版

可以计算得:

$$\begin{aligned}\frac{\omega_{12}}{du^1} &= \sqrt{\frac{E}{EG-F^2}}(-\frac{\partial}{\partial u^2}\sqrt{E}+\frac{\partial}{\partial u^1}\frac{F}{\sqrt{E}})\\ &= \frac{1}{2}\frac{1}{\sqrt{EG-F^2}}(-E_2+2F_1-\frac{F}{E}E_1)\\ &= \frac{\sqrt{EG-F^2}}{E}\Gamma^2_{11},\end{aligned}$$

$$\begin{aligned}\frac{\omega_{12}}{du^2} &= \frac{F}{E^2}\Gamma^2_{11}-\frac{1}{\sqrt{E}}\frac{\partial}{\partial u^1}(\sqrt{\frac{EG-F^2}{E}})\\ &= \frac{F}{E^2}\Gamma^2_{11}-\frac{1}{2}\frac{1}{\sqrt{EG-F^2}}(G_1-\frac{2FF_1}{E}+\frac{F^2E_1}{E})\\ &= \frac{\sqrt{EG-F^2}}{E}\Gamma^2_{12}.\end{aligned}$$

□

9. 取沿曲线 C 的与 $\mathbf{v}$ 垂直的单位向量场 $\mathbf{w}$，那么

$$\frac{D\mathbf{v}}{dt}=\mathbf{0}\Longleftrightarrow$$

$$\langle\frac{D\mathbf{v}}{dt},\mathbf{v}\rangle=0 \tag{1}$$

$$\langle\frac{D\mathbf{v}}{dt},\mathbf{w}\rangle=0. \tag{2}$$

由于 $0=d|\mathbf{v}|^2=\langle\frac{D\mathbf{v}}{dt},v\rangle$，等式 (1) 恒成立，故 $\mathbf{v}$ 沿曲线 C 平行当且仅当等式 (2) 成立。

$$\begin{aligned}\langle\frac{D\mathbf{v}}{dt},\mathbf{w}\rangle &= \frac{d}{dt}\Big(\frac{1}{\sqrt{E}}\Big)\langle\mathbf{r}_1,\mathbf{w}\rangle+\frac{1}{\sqrt{E}}\Gamma^\beta_{1\alpha}\frac{du^\alpha}{dt}\langle\mathbf{r}_\beta,\mathbf{w}\rangle\\ &= \frac{1}{\sqrt{E}}\langle\mathbf{r}_2,\mathbf{w}\rangle\Gamma^2_{1\alpha}\frac{du^\alpha}{dt},\end{aligned}$$

曲面参数应满足 $\langle\mathbf{r}_2,\mathbf{w}\rangle\neq0$，故等式 (2) 等价于

$$\Gamma^2_{1\alpha}\frac{du^\alpha}{dt}=0.$$

□

注：由于v=e_1，也可直接用T6的结论

10. (1) 球面第一基本形式

$$I = a^2(du)^2 + a^2\cos^2 u(dv)^2,$$

根据 Liouville 公式得到

$$k_g = \frac{d\theta}{ds} - \frac{1}{a}\frac{\sin u}{\cos u}\sin\theta.$$

又有

$$\sin\theta = \frac{1}{|\mathbf{r}_v|}\langle\frac{d}{ds}\mathbf{r}, \mathbf{r}_v\rangle = a\cos u\frac{dv}{ds},$$

代入即得.

(2) $\theta \equiv \frac{\pi}{2}, C : (u_0, \frac{1}{a\cos u_0}s)$，代入第一问结论得到

$$k_g = -\frac{1}{a}\tan u_0.$$

□

12. (1)

$$k_g = \langle\frac{D\mathbf{e}_1}{ds}, \mathbf{e}_2\rangle = \lambda_1,$$

$$k_n = \langle\frac{d\mathbf{e}_1}{ds}, \mathbf{e}_3\rangle = \lambda_2.$$

(2)

$$\begin{aligned}(\mathbf{e}_3, \dot{\mathbf{e}}_3, \mathbf{e}_1) &= (\mathbf{e}_1, \mathbf{e}_3, \dot{\mathbf{e}}_3)\\ &= \langle\mathbf{e}_1 \wedge \mathbf{e}_3, \dot{\mathbf{e}}_3\rangle\\ &= -\langle\mathbf{e}_2, \dot{\mathbf{e}}_3\rangle\\ &= \lambda_3.\end{aligned}$$

□

13.

$$\mathbf{w} = \mathbf{e}, \quad \mathbf{n}\wedge\mathbf{w} = \mathbf{e}_2,$$

$$\mathbf{v} = \tilde{\mathbf{e}}, \quad \mathbf{n}\wedge\mathbf{v} = \tilde{\mathbf{e}}_2,$$

有

$$\begin{cases}\tilde{\mathbf{e}_1} = \cos\theta\mathbf{e}_1 + \sin\theta\mathbf{e}_2,\\ \tilde{\mathbf{e}_2} = -\sin\theta\mathbf{e}_1 + \cos\theta\mathbf{e}_2.\end{cases}$$

$$\begin{aligned}\langle\frac{D\tilde{\mathbf{e}}_1}{dt},\tilde{\mathbf{e}}_2\rangle &= \frac{d\theta}{dt}+\cos^2\theta\langle\frac{D\mathbf{e}_1}{dt},\mathbf{e}_2\rangle-\sin^2\theta\langle\frac{D\mathbf{e}_2}{dt},\mathbf{e}_1\rangle\\ &= \frac{d\theta}{dt}+\langle\frac{D\mathbf{e}_1}{dt},\mathbf{e}_2\rangle.\end{aligned}$$

□

14. 旋转曲面的第一基本形式

$$I=(f'^2+g'^2)\ (du)^2+f^2(dv)^2,$$

根据 Liouville 公式

$$0=k_g=\frac{d\theta}{ds}+\frac{f'}{f\sqrt{f'^2+g'^2}}\sin\theta.$$

由于 $\cos\theta=\frac{1}{|\mathbf{r}_u|}\langle\frac{d\mathbf{r}}{ds},\mathbf{r}_u\rangle=\sqrt{f'^2+g'^2}\frac{du}{ds}$, 故有

$$f\cos\theta\frac{d\theta}{ds}+f'\frac{du}{ds}\sin\theta,$$

$$f(u)\sin\theta=\text{const}.$$

□